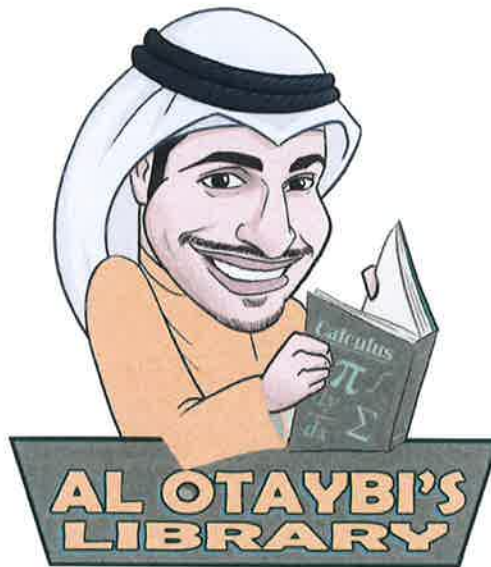


Kuwait University

2021/2022

Spring course



Calculus B

Final

Section : **8.1 - 8.2 - 8.3**

8.1 Arc Length.

8.2 Area of a Surface of Revolution.

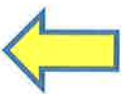
8.3 Applications to Physics and Engineering.

(15 a/b)

8.1 Arc length



$$y = f(x), \quad a \leq x \leq b$$



$$L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

$$x = f(y), \quad a \leq y \leq b$$



$$L = \int_a^b \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy$$



مراجعة Calculus A



لا يمكن اجراء التكامل على أي مطلق , فهند وجوده يجب

← إعادة تعريفه →

Examples

$$\int_2^5 |x+1| dx$$

$$\int_2^5 (x+1) dx$$

$$\begin{aligned} x+1 &= 0 \\ x &= -1 \\ &\notin (2, 5) \end{aligned}$$

$$\int_{-3}^{-1} |x+1| dx$$

$$\int_{-3}^{-1} (x+1) dx$$

$$\begin{aligned} x+1 &= 0 \\ x &= -1 \\ &\in (-3, -1) \end{aligned}$$

$$\int_{-3}^5 |x+1| dx$$

$$\int_{-3}^{-1} (x+1) dx + \int_{-1}^5 (x+1) dx$$

$$\begin{aligned} x+1 &= 0 \\ x &= -1 \\ &\in (-3, 5) \end{aligned}$$

Find the arc length of the curve

$$f(x) = \frac{1}{3}x^{\frac{3}{2}} - x^{\frac{1}{2}}, \quad 1 \leq x \leq 4$$

$$L = \int_1^4 \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{3} \cdot \frac{3}{2} x^{\frac{1}{2}} - \frac{1}{2} x^{-\frac{1}{2}} = \frac{x^{\frac{1}{2}}}{2} - \frac{1}{2x^{\frac{1}{2}}}$$

$$= \frac{x^{\frac{1}{2}} x^{\frac{1}{2}}}{2x^{\frac{1}{2}}} - \frac{1}{2x^{\frac{1}{2}}} = \frac{x-1}{2x^{\frac{1}{2}}}$$

$$\Rightarrow \left(\frac{dy}{dx}\right)^2 = \left(\frac{x-1}{2x^{\frac{1}{2}}}\right)^2 = \frac{(x-1)^2}{(2x^{\frac{1}{2}})^2} = \frac{x^2-2x+1}{4x}$$

$$\Rightarrow 1 + \left(\frac{dy}{dx}\right)^2 = 1 + \frac{x^2-2x+1}{4x} = \frac{4x}{4x} + \frac{x^2-2x+1}{4x}$$

$$= \frac{4x + x^2 - 2x + 1}{4x} = \frac{x^2 + 2x + 1}{4x}$$

$$L = \int_1^4 \sqrt{\frac{x^2+2x+1}{4x}} dx = \int_1^4 \frac{\sqrt{x^2+2x+1}}{\sqrt{4x}} dx$$

$$= \int_1^4 \frac{\sqrt{(x+1)^2}}{2\sqrt{x}} dx \quad \begin{matrix} x+1 \\ x \neq -1 \end{matrix}$$

$x^2+2x+1 = (x+1)^2 - 1 + 1 = (x+1)^2$

$$= \frac{1}{2} \int_1^4 \frac{x+1}{x^{\frac{1}{2}}} dx = \frac{1}{2} \int_1^4 \left(\frac{x}{x^{\frac{1}{2}}} + \frac{1}{x^{\frac{1}{2}}} \right) dx$$

$$= \frac{1}{2} \int_1^4 (x^{\frac{1}{2}} + x^{-\frac{1}{2}}) dx = \frac{1}{2} \left[\frac{2}{3} x^{\frac{3}{2}} + 2x^{\frac{1}{2}} \right]_1^4$$

$$= \frac{1}{2} \left[\left(\frac{2}{3} (4)^{\frac{3}{2}} + 2(4)^{\frac{1}{2}} \right) - \left(\frac{2}{3} + 2 \right) \right] = \frac{1}{2} \left(\frac{16}{3} + 4 - \frac{2}{3} - 2 \right)$$

Find the length of the arc of

$$y = \ln(\sec x) \quad , \quad x \in \left[0, \frac{\pi}{4}\right]$$

$$L = \int_0^{\frac{\pi}{4}} \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

$$\Rightarrow \frac{dy}{dx} = \frac{\sec x \tan x}{\sec x} = \tan x$$

$$\Rightarrow \left(\frac{dy}{dx}\right)^2 = \tan^2 x$$

$$\Rightarrow 1 + \left(\frac{dy}{dx}\right)^2 = 1 + \tan^2 x = \sec^2 x$$

$$L = \int_0^{\frac{\pi}{4}} \sqrt{\sec^2 x} dx$$

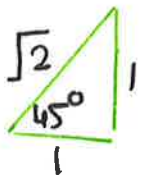
$\searrow$
 $|\sec x|$

$$= \int_0^{\frac{\pi}{4}} \sec x dx$$
$$= \left[\ln |\sec x + \tan x| \right]_0^{\frac{\pi}{4}}$$

$$= \ln |\sqrt{2} + 1| - \ln |1 + 0|$$

$$= \ln |\sqrt{2} + 1|$$

s	A [#]
t	c



Find the length of the arc of

$$y = \ln \sqrt{\tanh(x)} \quad , \quad 1 \leq x \leq 2$$

$$L = \int_1^2 \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

$$y = \ln [\tanh x]^{\frac{1}{2}} = \frac{1}{2} \ln \tanh x$$

$$\begin{aligned} \Rightarrow \frac{dy}{dx} &= \frac{1}{2} \frac{\operatorname{sech}^2 x}{\tanh x} = \frac{1}{2} \frac{\frac{1}{\cosh^2 x}}{\frac{\sinh x}{\cosh x}} \\ &= \frac{1}{2 \sinh x \cosh x} = \frac{1}{\sinh 2x} = \operatorname{csch} 2x \end{aligned}$$

$$\Rightarrow \left(\frac{dy}{dx}\right)^2 = \operatorname{csch}^2 2x$$

$$\Rightarrow 1 + \left(\frac{dy}{dx}\right)^2 = 1 + \operatorname{csch}^2 2x = \operatorname{coth}^2 2x$$

$$\begin{aligned} L &= \int_1^2 \sqrt{\operatorname{coth}^2 2x} dx \\ &= \int_1^2 \operatorname{coth} 2x dx \\ &= \int_1^2 \frac{\cosh 2x}{\sinh 2x} dx = \frac{1}{2} \int_1^2 \frac{\cosh 2x}{\sinh 2x} dx \\ &= \frac{1}{2} [\ln |\sinh 2x|]_1^2 \\ &= \frac{1}{2} [\ln |\sinh 4| - \ln |\sinh 2|] \end{aligned}$$

Find the arc length of the curve

$$x = \ln(\cos y) \quad , \quad 0 \leq y \leq \frac{\pi}{6}$$

$$L = \int_0^{\frac{\pi}{6}} \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy$$

$$\Rightarrow \frac{dx}{dy} = \frac{-\sin y}{\cos y} = -\tan y$$

$$\Rightarrow \left(\frac{dx}{dy}\right)^2 = (-\tan y)^2 = \tan^2 y$$

$$\Rightarrow 1 + \left(\frac{dx}{dy}\right)^2 = 1 + \tan^2 y = \sec^2 y$$

$$L = \int_0^{\frac{\pi}{6}} \sqrt{\sec^2 y} dy$$

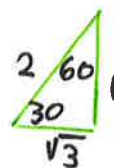
$$= \int_0^{\frac{\pi}{6}} \sec y dy$$

$$= \left[\ln |\sec y + \tan y| \right]_0^{\frac{\pi}{6}}$$

$$= \ln \left| \frac{2}{\sqrt{3}} + \frac{1}{\sqrt{3}} \right| - \ln |1 + 0|$$

$$= \ln \frac{3}{\sqrt{3}}$$

S	A	#
t	c	



$$\text{If } g(y) = \frac{1}{3}(y^2 + 2)^{\frac{3}{2}}$$

Find the length of the graph of g from $y = 0$ to $y = \sqrt{3}$

$$L = \int_0^{\sqrt{3}} \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy$$

$$\Rightarrow \frac{dx}{dy} = \frac{1}{3} \cdot \frac{3}{2} (y^2 + 2)^{\frac{1}{2}} \cdot 2y = y(y^2 + 2)^{\frac{1}{2}}$$

$$\Rightarrow \left(\frac{dx}{dy}\right)^2 = \left[y(y^2 + 2)^{\frac{1}{2}}\right]^2 = y^2(y^2 + 2) = y^4 + 2y^2$$

$$\Rightarrow 1 + \left(\frac{dx}{dy}\right)^2 = 1 + y^4 + 2y^2 = y^4 + 2y^2 + 1$$

$$L = \int_0^{\sqrt{3}} \sqrt{y^4 + 2y^2 + 1} dy$$

$$= \int_0^{\sqrt{3}} \sqrt{(y^2 + 1)^2} dy$$

$$= \int_0^{\sqrt{3}} (y^2 + 1) dy$$

$$= \left[\frac{y^3}{3} + y \right]_0^{\sqrt{3}}$$

$$= \frac{(\sqrt{3})^3}{3} + \sqrt{3} - 0$$

Find the arc length of the curve

C: $y = \frac{2}{3}x^{\frac{3}{2}}$ between the points $(0, 0)$ and $(3, 2\sqrt{3})$

$$L = \int_0^3 \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

$$\Rightarrow \frac{dy}{dx} = \frac{2}{3} \cdot \frac{3}{2} x^{\frac{1}{2}} = x^{\frac{1}{2}}$$

$$\Rightarrow \left(\frac{dy}{dx}\right)^2 = \left(x^{\frac{1}{2}}\right)^2 = x$$

$$\Rightarrow 1 + \left(\frac{dy}{dx}\right)^2 = 1 + x$$

$$L = \int_0^3 \sqrt{1+x} dx$$

$$= \int_0^3 (1+x)^{\frac{1}{2}} dx$$

$$= \left[\frac{(1+x)^{\frac{3}{2}}}{\frac{3}{2}} \right]_0^3$$

$$= \frac{2}{3} \left[(4)^{\frac{3}{2}} - (1)^{\frac{3}{2}} \right]$$

$$= \frac{2}{3} [8 - 1]$$

$$= \frac{2}{3} [7] = \frac{14}{3}$$



Tricky question !!



Find the arc length of the curve

$$y = \int_1^x \sqrt{t^2 - 1} dt, \quad 1 \leq x \leq 4$$

$$L = \int_1^4 \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

$$\Rightarrow \frac{dy}{dx} = \sqrt{x^2 - 1} (1) - 0 = \sqrt{x^2 - 1}$$

$$\Rightarrow \left(\frac{dy}{dx}\right)^2 = (\sqrt{x^2 - 1})^2 = x^2 - 1$$

$$\Rightarrow 1 + \left(\frac{dy}{dx}\right)^2 = 1 + x^2 - 1 = x^2$$

$$L = \int_1^4 \sqrt{x^2} dx$$

$\rightarrow |x| \quad x=0 \notin (1,4)$

$$= \int_1^4 x dx$$

$$= \left(\frac{x^2}{2}\right)_1^4 = \frac{16}{2} - \frac{1}{2} = \frac{15}{2}$$



Arc length function



Find the arc length function for the curve

$$y = \sin^{-1}x + \sqrt{1-x^2}$$

with starting point $(0, 1)$.

$$L = \int_0^x \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

$$\begin{aligned} \Rightarrow \frac{dy}{dx} &= \frac{1}{\sqrt{1-x^2}} + \frac{-2x}{2\sqrt{1-x^2}} \\ &= \frac{1}{\sqrt{1-x^2}} - \frac{x}{\sqrt{1-x^2}} = \frac{1-x}{\sqrt{1-x^2}} \end{aligned}$$

$$\begin{aligned} \Rightarrow \left(\frac{dy}{dx}\right)^2 &= \left(\frac{1-x}{\sqrt{1-x^2}}\right)^2 = \frac{(1-x)^2}{1-x^2} = \frac{1-2x+x^2}{1-x^2} \\ &= \frac{x^2-2x+1}{1-x^2} = \frac{(x-1)(x-1)}{(1-x)(1+x)} = \frac{-(1-x)(x-1)}{(1-x)(1+x)} = \frac{1-x}{1+x} \end{aligned}$$

$$\Rightarrow 1 + \left(\frac{dy}{dx}\right)^2 = 1 + \frac{1-x}{1+x} = \frac{1+x}{1+x} + \frac{1-x}{1+x} = \frac{2}{1+x}$$

$$\begin{aligned} L &= \int_0^x \sqrt{\frac{2}{1+x}} dx = \int_0^x \frac{\sqrt{2}}{\sqrt{1+x}} dx \\ &= \sqrt{2} \int_0^x (1+x)^{-\frac{1}{2}} dx = \sqrt{2} \left[\frac{(1+x)^{\frac{1}{2}}}{\frac{1}{2}} \right]_0^x \\ &= \sqrt{2} \cdot 2 \cdot \left[(1+x)^{\frac{1}{2}} - 1 \right] \end{aligned}$$



Multiple choice



Select the integral that represents the length of the curve

$$y = \sin x, 0 \leq x \leq \pi/2$$

(A) $\int_0^1 \sqrt{1 + \cos^2 x} dx$

(b) $\int_0^{\pi/2} \sqrt{1 + \cos^2 x} dx$

(c) $\int_0^{\pi/2} 2\pi x \sqrt{1 + \cos^2 x} dx$

(d) $\int_0^1 \sqrt{1 + \sin^2 x} dx$

May 2017

$$L = \int_0^{\pi/2} \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

$$\frac{dy}{dx} = \cos x$$

$$\left(\frac{dy}{dx}\right)^2 = \cos^2 x$$

$$\Rightarrow 1 + \left(\frac{dy}{dx}\right)^2 = 1 + \cos^2 x$$

$$L = \int_0^{\pi/2} \sqrt{1 + \cos^2 x} dx$$

(b)

8.2 Surface Area



$$y = f(x), \quad a \leq x \leq b$$

about x - axis

about y - axis

$$x = f(y), \quad a \leq y \leq b$$

about x - axis

about y - axis

$$S = 2\pi \int_a^b f(x) \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

$$S = 2\pi \int_a^b x \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

$$S = 2\pi \int_a^b y \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy$$

$$S = 2\pi \int_a^b f(y) \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy$$

Find the area of the surface generated by revolving the curve

$$y = 2\sqrt{x} \quad , \quad 1 \leq x \leq 2 \quad , \quad \text{about } x - \text{axis}$$

$$S = 2\pi \int_1^2 y \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

$$\Rightarrow \frac{dy}{dx} = 2 \cdot \frac{1}{2\sqrt{x}} = \frac{1}{\sqrt{x}} \quad \text{غير ابدل}$$

$$\Rightarrow \left(\frac{dy}{dx}\right)^2 = \left(\frac{1}{\sqrt{x}}\right)^2 = \frac{1}{x}$$

$$\Rightarrow 1 + \left(\frac{dy}{dx}\right)^2 = 1 + \frac{1}{x} = \frac{x}{x} + \frac{1}{x} = \frac{x+1}{x}$$

$$S = 2\pi \int_1^2 2\sqrt{x} \sqrt{\frac{x+1}{x}} dx$$

$$= 4\pi \int_1^2 \cancel{\sqrt{x}} \frac{\sqrt{x+1}}{\cancel{\sqrt{x}}} dx$$

$$= 4\pi \int_1^2 (x+1)^{\frac{1}{2}} dx$$

$$= 4\pi \left[\frac{(x+1)^{\frac{3}{2}}}{\frac{3}{2}} \right]_1^2 = 4\pi \frac{2}{3} \left[(x+1)^{\frac{3}{2}} \right]_1^2$$

$$= \frac{8\pi}{3} \left[3^{\frac{3}{2}} - 2^{\frac{3}{2}} \right]$$

Find the area of the surface generated by revolving the curve

$$y = x^2 \quad , \quad 1 \leq x \leq 2 \quad , \quad \text{about } y - \text{axis}$$

$$S = 2\pi \int_1^2 x \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

← *هذا*

$$\Rightarrow \frac{dy}{dx} = 2x$$

$$\Rightarrow \left(\frac{dy}{dx}\right)^2 = (2x)^2 = 4x^2$$

$$\Rightarrow 1 + \left(\frac{dy}{dx}\right)^2 = 1 + 4x^2$$

$$S = 2\pi \int_1^2 x \sqrt{1 + 4x^2} dx$$

$$= 2\pi \int_1^2 x (1 + 4x^2)^{\frac{1}{2}} dx$$

Let $u = 1 + 4x^2$

$$du = 8x dx \rightarrow \frac{1}{8} du = x dx$$

$$= 2\pi \frac{1}{8} \int u^{\frac{1}{2}} du = -$$

$$= \frac{\pi}{4} \frac{u^{\frac{3}{2}}}{\frac{3}{2}} = \frac{\pi}{4} \frac{2}{3} \left[(1 + 4x^2)^{\frac{3}{2}} \right]_1^2$$

$$= \frac{\pi}{6} \left[(17)^{\frac{3}{2}} - (5)^{\frac{3}{2}} \right]$$

$$= \frac{\pi}{6} \left[(17)^{\frac{3}{2}} - (5)^{\frac{3}{2}} \right]$$

Find the area of the surface obtained by revolving the curve

$y = \sqrt{x+1}$, $1 \leq x \leq 5$ about the x - axis

$$S = 2\pi \int_1^5 y \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{2\sqrt{x+1}}$$

$$\Rightarrow \left(\frac{dy}{dx}\right)^2 = \left(\frac{1}{2\sqrt{x+1}}\right)^2 = \frac{1}{4(x+1)}$$

$$\begin{aligned} \Rightarrow 1 + \left(\frac{dy}{dx}\right)^2 &= 1 + \frac{1}{4(x+1)} = \frac{4(x+1)}{4(x+1)} + \frac{1}{4(x+1)} \\ &= \frac{4x+4+1}{4(x+1)} = \frac{4x+5}{4(x+1)} \end{aligned}$$

$$S = 2\pi \int_1^5 \sqrt{x+1} \sqrt{\frac{4x+5}{4(x+1)}} dx$$

$$= 2\pi \int_1^5 \cancel{\sqrt{x+1}} \frac{\sqrt{4x+5}}{\cancel{\sqrt{4}} \cancel{\sqrt{x+1}}} dx$$

$$= \pi \int_1^5 (4x+5)^{\frac{1}{2}} dx$$

$$= \pi \left[\frac{(4x+5)^{\frac{3}{2}}}{\frac{2 \cdot 4}{2} \cdot \frac{3}{2}} \right]_1^5 = \pi \left[\frac{(4x+5)^{\frac{3}{2}}}{6} \right]_1^5$$

$$= \frac{\pi}{6} \left[(25)^{\frac{3}{2}} - (9)^{\frac{3}{2}} \right] = \frac{\pi}{6} \left[125 - 27 \right] = \frac{98\pi}{6}$$

Find the area of the surface generated by rotating the curve

قراءة

$x = \sqrt{8y}$, $y \in [0, 1]$ about y -axis

$$S = 2\pi \int_0^1 x \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy$$

$$\Rightarrow \frac{dx}{dy} = \frac{8}{2\sqrt{8y}} = \frac{4}{\sqrt{8y}}$$

$$\Rightarrow \left(\frac{dx}{dy}\right)^2 = \left(\frac{4}{\sqrt{8y}}\right)^2 = \frac{16}{8y} = \frac{2}{y}$$

$$\Rightarrow 1 + \left(\frac{dx}{dy}\right)^2 = 1 + \frac{2}{y} = \frac{y}{y} + \frac{2}{y} = \frac{y+2}{y}$$

$$S = 2\pi \int_0^1 \sqrt{8y} \sqrt{\frac{y+2}{y}} dy$$

$$= 2\pi \int_0^1 \sqrt{8} \sqrt{y} \frac{\sqrt{y+2}}{\sqrt{y}} dy$$

$$= 2\pi \sqrt{8} \int_0^1 (y+2)^{\frac{1}{2}} dy$$

$$= 2\pi \sqrt{8} \left[\frac{(y+2)^{\frac{3}{2}}}{\frac{3}{2}} \right]_0^1$$

$$= 2\pi \sqrt{8} \cdot \frac{2}{3} \left[3^{\frac{3}{2}} - 2^{\frac{3}{2}} \right]$$

Find the area of the surface generated by revolving the curve

$$y = \cosh x \quad , \quad x \in [0, 1]$$

a) around y - axis

b) around x - axis

$$a) S = 2\pi \int_0^1 x \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

تفاضل

$$\Rightarrow \frac{dy}{dx} = \sinh x$$

$$\Rightarrow \left(\frac{dy}{dx}\right)^2 = \sinh^2 x$$

$$\Rightarrow 1 + \left(\frac{dy}{dx}\right)^2 = 1 + \sinh^2 x = \cosh^2 x$$

$$S = 2\pi \int_0^1 x \sqrt{\cosh^2 x} dx$$

$\sqrt{\cosh^2 x} = |\cosh x|$

$$= 2\pi \int_0^1 x \cosh x dx$$

$$u = x \quad \begin{matrix} \nearrow \\ \searrow \end{matrix} \quad \begin{matrix} dv = \cosh x dx \\ v = \sinh x \end{matrix}$$
$$du = dx$$

$$= 2\pi \left[x \sinh x - \int \sinh x dx \right]_0^1$$

$$= 2\pi \left[x \sinh x - \cosh x \right]_0^1$$

$$= 2\pi \left[(\sinh(1) - \cosh(1)) - (0 - 1) \right]$$

$$b) \quad S = 2\pi \int_0^1 y \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

غير ← ابدل

From @ $\Rightarrow 1 + \left(\frac{dy}{dx}\right)^2 = \cosh^2 x$

$$S = 2\pi \int_0^1 \cosh x \sqrt{\cosh^2 x} dx$$

$\rightarrow |\cosh x|$

$$= 2\pi \int_0^1 \cosh x \cosh x dx$$

$$= 2\pi \int_0^1 \cosh^2 x dx$$

$$= 2\pi \cdot \frac{1}{2} \int_0^1 (1 + \cosh 2x) dx$$

$$= \pi \left[x + \frac{\sinh 2x}{2} \right]_0^1$$

$$= \pi \left[\left(1 + \frac{\sinh 2}{2}\right) - (0+0) \right]$$

$$= \pi \left(1 + \frac{\sinh 2}{2}\right)$$

Let the curve $C: y = \frac{1}{8}x^2 - \ln x$, for $x \in [1, 2]$

Find the area of the surface obtained by revolving the curve C about y -axis.

$$S = 2\pi \int_1^2 x \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{8} \cdot 2x - \frac{1}{x} = \frac{x}{4} - \frac{1}{x} = \frac{x \cdot x - 4}{4x} = \frac{x^2 - 4}{4x}$$

$$\Rightarrow \left(\frac{dy}{dx}\right)^2 = \left(\frac{x^2 - 4}{4x}\right)^2 = \frac{(x^2 - 4)^2}{(4x)^2} = \frac{x^4 - 8x^2 + 16}{16x^2}$$

$$\Rightarrow 1 + \left(\frac{dy}{dx}\right)^2 = 1 + \frac{x^4 - 8x^2 + 16}{16x^2} = \frac{16x^2}{16x^2} + \frac{x^4 - 8x^2 + 16}{16x^2} = \frac{16x^2 + x^4 - 8x^2 + 16}{16x^2} = \frac{x^4 + 8x^2 + 16}{16x^2}$$

$$S = 2\pi \int_1^2 x \sqrt{\frac{x^4 + 8x^2 + 16}{16x^2}} dx$$

$$= 2\pi \int_1^2 x \frac{\sqrt{x^4 + 8x^2 + 16}}{\sqrt{16x^2}} dx$$

$\begin{aligned} & \frac{x^4 + 8x^2 + 16}{(x^2 + 4)^2} = \frac{(x^2 + 4)^2}{(x^2 + 4)^2} \\ & \frac{x^4 + 8x^2 + 16}{(x^2 + 4)^2} \end{aligned}$

$$= 2\pi \int_1^2 x \frac{\sqrt{(x^2 + 4)^2}}{4x} dx$$

$$= \frac{2\pi}{4} \int_1^2 (x^2 + 4) dx = \frac{\pi}{2} \left[\frac{x^3}{3} + 4x \right]_1^2$$

$$= \frac{\pi}{2} \left[\left(\frac{8}{3} + 8\right) - \left(\frac{1}{3} + 4\right) \right] = \frac{\pi}{2} \left[\frac{32}{3} - \frac{13}{3} \right] = \frac{19\pi}{6}$$

Find the area of the surface that is generated by revolving the circle

$y = \sqrt{9 - x^2}$, between $(0, 3)$ and $(2, \sqrt{5})$, about the y -axis

Fall 2018

$$S = 2\pi \int_0^2 x \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

$$\frac{dy}{dx} = \frac{-2x}{2\sqrt{9-x^2}} \longrightarrow \left(\frac{dy}{dx}\right)^2 = \frac{x^2}{9-x^2}$$

$$1 + \left(\frac{dy}{dx}\right)^2 = 1 + \frac{x^2}{9-x^2} = \frac{9-x^2}{9-x^2} + \frac{x^2}{9-x^2} = \frac{9}{9-x^2}$$

$$S = 2\pi \int_0^2 x \sqrt{\frac{9}{9-x^2}} dx$$

$$= 2\pi \int_0^2 x \frac{3}{\sqrt{9-x^2}} dx$$

$$= 6\pi \int_0^2 x(9-x^2)^{-\frac{1}{2}} dx$$

Let $u = 9 - x^2$

$$du = -2x dx \longrightarrow \frac{1}{2} du = -x dx$$

$$= 6\pi \left(\frac{1}{2}\right) \int u^{-\frac{1}{2}} du$$

$$= -3\pi \frac{u^{\frac{1}{2}}}{\frac{1}{2}} = -6\pi \left[(9-x^2)^{\frac{1}{2}}\right]_0^2$$

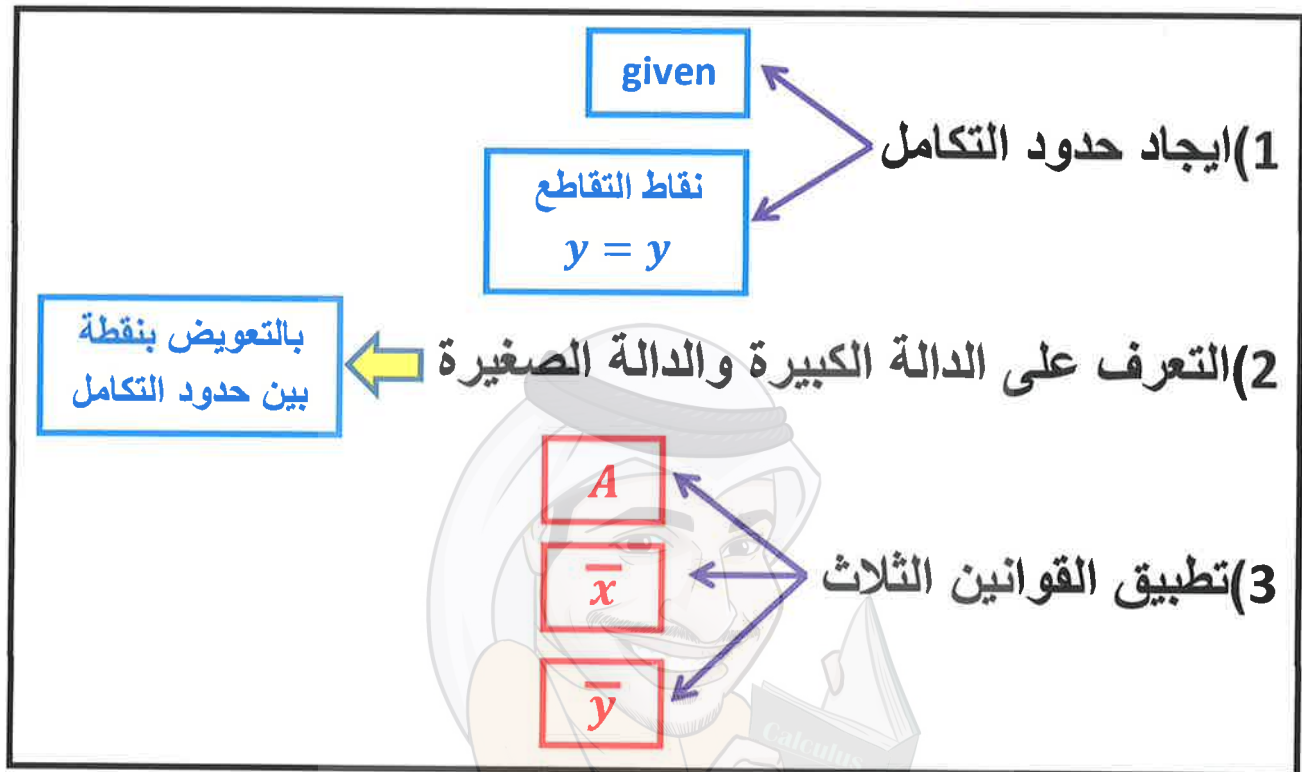
$$= -6\pi \left[5^{\frac{1}{2}} - (9)^{\frac{1}{2}}\right] = -6\pi [5^{\frac{1}{2}} - 3]$$



8.3 Centroid (center of mass)



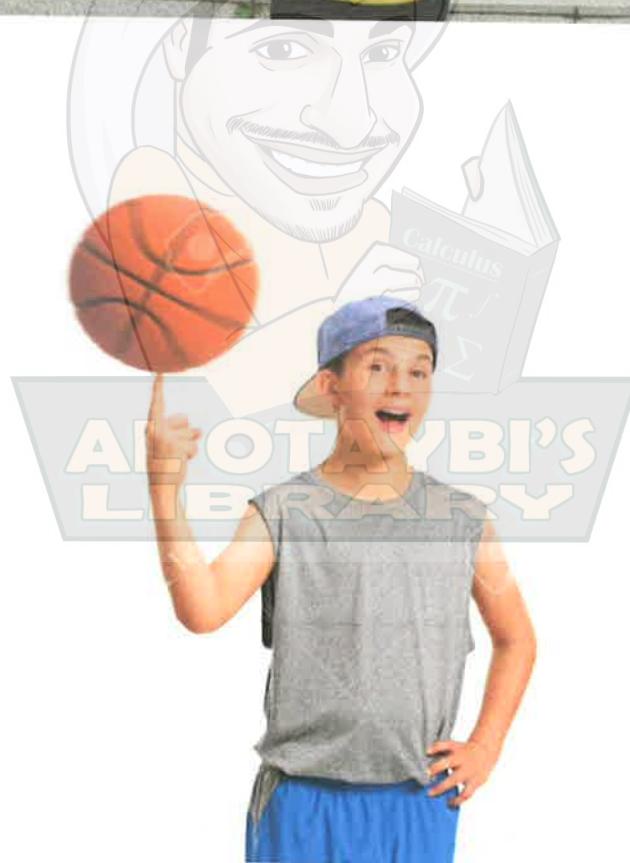
خطوات الحل



$$A = \int_a^b [(الدالة الكبيرة) - (الدالة الصغيرة)] dx$$

$$\bar{x} = \frac{1}{A} \int_a^b x [f(x) - g(x)] dx$$

$$\bar{y} = \frac{1}{2A} \int_a^b [f^2(x) - g^2(x)] dx$$



Find the centroid of the region bounded by the curves

$$y = \sec^2 x, \quad y = 0, \quad x = 0 \quad \text{and} \quad x = \frac{\pi}{4}$$

$$\begin{aligned} A &= \int_0^{\frac{\pi}{4}} (\sec^2 x - 0) dx \\ &= \left[\tan x \right]_0^{\frac{\pi}{4}} = \tan \frac{\pi}{4} - \tan 0 = 1 - 0 = 1 \end{aligned}$$

$$\begin{aligned} \bar{X} &= \frac{1}{A} \int_0^{\frac{\pi}{4}} x (\sec^2 x - 0) dx \\ &= \frac{1}{1} \int_0^{\frac{\pi}{4}} x \sec^2 x dx \end{aligned}$$

$u = x \quad \quad dv = \sec^2 x dx$
 $du = dx \quad \quad v = \tan x$

$$\begin{aligned} &\left[x \tan x \right]_0^{\frac{\pi}{4}} - \int_0^{\frac{\pi}{4}} \tan x dx \\ &\left[\frac{\pi}{4} \tan \frac{\pi}{4} - 0 \right] - \left[\ln |\sec x| \right]_0^{\frac{\pi}{4}} \\ &\frac{\pi}{4} (1) - [\ln \sqrt{2} - \ln 1] \\ &\frac{\pi}{4} - \ln \sqrt{2} \end{aligned}$$

$$\begin{aligned}
 \bar{y} &= \frac{1}{2A} \int_0^{\pi/2} \left[(\sec^2 x)^2 - (0)^2 \right] dx \\
 &= \frac{1}{2(1)} \int_0^{\pi/2} \sec^4 x \, dx \\
 &= \frac{1}{2} \int_0^{\pi/2} \sec^2 x \sec^2 x \, dx \\
 &= \frac{1}{2} \int_0^{\pi/2} (1 + \tan^2 x) \sec^2 x \, dx
 \end{aligned}$$

Let $u = \tan x$
 $du = \sec^2 x \, dx$

$$\begin{aligned}
 &= \frac{1}{2} \int (1 + u^2) \, du \\
 &= \frac{1}{2} \left[u + \frac{u^3}{3} \right] \\
 &= \frac{1}{2} \left[\tan x + \frac{(\tan x)^3}{3} \right]_0^{\pi/2} \\
 &= \frac{1}{2} \left[\left(1 + \frac{1}{3} \right) - 0 \right] \\
 &= \frac{1}{2} \cdot \frac{4}{3} = \frac{2}{3}
 \end{aligned}$$

$$\Rightarrow P(\bar{x}, \bar{y}) = \left(\frac{\pi}{4} - \ln \sqrt{2}, \frac{2}{3} \right)$$

Find the centroid of the region bounded by

$y = 3 - 2x$ and the parabola $y = 6 - x^2$

$$y_1 = 3 - 2x \quad y_2 = 6 - x^2$$

$$y = y$$

$$3 - 2x = 6 - x^2$$

$$x^2 - 6 + 3 - 2x = 0$$

$$x^2 - 2x - 3 = 0$$

$$(x + 1)(x - 3) = 0$$

$$x = -1 \quad x = 3$$

at $x=0$ $\begin{cases} y_1 = 3 - 0 = 3 \\ y_2 = 6 - 0 = 6 \end{cases} \Rightarrow y_2 > y_1$

$$A = \int_{-1}^3 (y_2 - y_1) dx$$
$$= \int_{-1}^3 (6 - x^2 - (3 - 2x)) dx$$

$$= \int_{-1}^3 (6 - x^2 - 3 + 2x) dx$$
$$= \int_{-1}^3 (3 - x^2 + 2x) dx = \left[3x - \frac{x^3}{3} + \frac{2x^2}{2} \right]_{-1}^3$$

$$= (9 - \frac{27}{3} + 9) - (-3 + \frac{1}{3} + 1) = 9 - (-2 + \frac{1}{3})$$
$$= 9 - (-\frac{5}{3}) + \frac{27}{3} + \frac{5}{3} = \frac{32}{3}$$

$$\bar{X} = \frac{1}{A} \int_{-1}^3 X (y_2 - y_1) dx$$

$$= \frac{1}{\frac{32}{3}} \int_{-1}^3 X (6 - x^2 - (3 - 2x)) dx$$

$$= \frac{3}{32} \int_{-1}^3 X (6 - x^2 - 3 + 2x) dx$$

$6 - x^2 - 3 + 2x = 3 - x^2 + 2x$

$$= \frac{3}{32} \int_{-1}^3 (3x - x^3 + 2x^2) dx$$

$$= \frac{3}{32} \left[\frac{3x^2}{2} - \frac{x^4}{4} + \frac{2x^3}{3} \right]_{-1}^3$$

$$= \frac{3}{32} \left[\left(\frac{3(9)}{2} - \frac{81}{4} + \frac{2(27)}{3} \right) - \left(\frac{3}{2} - \frac{1}{4} - \frac{2}{3} \right) \right]$$

$$\bar{y} = \frac{1}{2A} \int_{-1}^3 [y_2^2 - y_1^2] dx$$

$$= \frac{1}{2 \cdot \frac{32}{3}} \int_{-1}^3 [(6 - x^2)^2 - (3 - 2x)^2] dx$$

$$36 - 12x^2 + x^4 - (9 - 12x + 4x^2)$$

$$36 - 12x^2 + x^4 - 9 + 12x - 4x^2$$

$$27 - 16x^2 + 12x + x^4$$

$$= \frac{3}{64} \int_{-1}^3 [27 - 16x^2 + 12x + x^4] dx$$

$$= \frac{3}{64} \left[27x - \frac{16x^3}{3} + \frac{12x^2}{2} + \frac{x^5}{5} \right]_{-1}^3$$

$$= \frac{3}{64} \left[\left(27(3) - \frac{16(27)}{3} + 6(9) + \frac{(3)^5}{5} \right) - \left(-27 + \frac{16}{3} + 6 - \frac{1}{5} \right) \right]$$

Find the centroid of the region bounded by

Fall 2017

$$y = \sin(2x) \text{ and } y = 0 \text{ for } 0 \leq x \leq \frac{\pi}{2}$$

$$A = \int_0^{\frac{\pi}{2}} [\sin(2x) - 0] dx = \left[-\frac{\cos(2x)}{2} \right]_0^{\frac{\pi}{2}} = -\frac{\cos(\pi)}{2} + \frac{1}{2} = \frac{1}{2} + \frac{1}{2} = \boxed{1}$$

$$\bar{x} = \frac{1}{A} \int_0^{\frac{\pi}{2}} x [\sin(2x) - 0] dx$$

① $u = x \quad dv = \sin(2x)$
 $du = dx \quad v = -\frac{\cos(2x)}{2}$

$$= \left[-\frac{1}{2} x \cos(2x) \right]_0^{\frac{\pi}{2}} + \frac{1}{2} \int_0^{\frac{\pi}{2}} \cos 2x dx$$

$$= \left[-\frac{1}{2} x \cos(2x) + \frac{1}{4} \sin(2x) \right]_0^{\frac{\pi}{2}}$$

$$= \left(-\frac{1}{2} \cdot \frac{\pi}{2} \cdot (-1) + 0 \right) - (0) = \boxed{\frac{\pi}{4}}$$

$$\bar{y} = \frac{1}{2A} \int_0^{\frac{\pi}{2}} [\sin^2(2x) - 0^2] dx$$

$$= \frac{1}{2(1)} \cdot \frac{1}{2} \int_0^{\frac{\pi}{2}} (1 - \cos(4x)) dx$$

$$= \frac{1}{4} \left[x - \frac{\sin(4x)}{4} \right]_0^{\frac{\pi}{2}}$$

$$= \frac{1}{4} \left[\left(\frac{\pi}{2} - 0 \right) - 0 \right] = \boxed{\frac{\pi}{8}}$$

$$\Rightarrow P(\bar{x}, \bar{y}) = P\left(\frac{\pi}{4}, \frac{\pi}{8}\right)$$

Find the centroid of the region bounded by the curves

$$y = x^3, \quad x + y = 2 \quad \text{and} \quad x = 0$$

لازم أظلم أن
بروحها على شان
نعتبر دالة

مطيبي حد واحد
لازم أظلم الثاني

$$y_1 = x^3$$

$$y_2 = 2 - x$$

$$x = 0$$

$$y = y$$

$$x^3 = 2 - x$$

$$x^3 + x - 2 = 0$$

لا يمكن تبليها !!

By guessing

$$\text{at } x = 1$$

$$\rightarrow \text{L.H.S} = (1)^3 + (1) - 2 = 0 = \text{R.H.S}$$

$$\Rightarrow \therefore x = 1$$

$$\text{at } x = \frac{1}{2}$$

$$y_1 = \left(\frac{1}{2}\right)^3 = \frac{1}{8}$$

$$y_2 = 2 - \frac{1}{2} = \frac{3}{2}$$

$$\Rightarrow \therefore y_2 > y_1$$

$$A = \int_0^1 (y_2 - y_1) dx$$

$$= \int_0^1 (2 - x - x^3) dx$$

$$= \left[2x - \frac{x^2}{2} - \frac{x^4}{4} \right]_0^1$$

$$= 2 - \frac{1}{2} - \frac{1}{4} - 0 = \frac{8}{4} - \frac{2}{4} - \frac{1}{4} = \frac{5}{4}$$

$$\bar{x} = \frac{1}{A} \int_0^1 x [y_2 - y_1] dx$$

$$= \frac{1}{\frac{5}{2}} \int_0^1 x [2 - x - x^3] dx$$

$$= \frac{2}{5} \int_0^1 [2x - x^2 - x^4] dx$$

$$= \frac{2}{5} \left[\frac{2x^2}{2} - \frac{x^3}{3} - \frac{x^5}{5} \right]_0^1$$

$$= \frac{2}{5} \left[1 - \frac{1}{3} - \frac{1}{5} - 0 \right]$$

$$= \frac{2}{5} \left[\frac{15}{15} - \frac{5}{15} - \frac{3}{15} \right] = \frac{2}{5} \cdot \frac{7}{15} = \frac{28}{75}$$

$$\bar{y} = \frac{1}{2A} \int_0^1 [y_2^2 - y_1^2] dx$$

$$= \frac{1}{2 \cdot \frac{5}{2}} \int_0^1 [(2-x)^2 - (x^3)^2] dx$$

$$= \frac{1}{5} \int_0^1 [4 - 4x + x^2 - x^6] dx$$

$$= \frac{2}{5} \left[4x - \frac{4x^2}{2} + \frac{x^3}{3} - \frac{x^7}{7} \right]_0^1$$

$$= \frac{2}{5} \left[4 - 2 + \frac{1}{3} - \frac{1}{7} \right] = \dots = \frac{92}{105}$$

$$\Rightarrow P(\bar{x}, \bar{y}) = P\left(\frac{28}{75}, \frac{92}{105}\right)$$

Find the centroid of the region in the first quadrant that is bounded by the graph $y = e^x$ and the line $x = 1$.

Jan 2017

$$y = e^x \quad x = 1 \quad y = 0 \quad x = 0$$

$$A = \int_0^1 (e^x - 0) dx = [e^x]_0^1 = \boxed{e - 1}$$

$$\bar{x} = \frac{1}{A} \int_0^1 x(e^x - 0) dx$$

$$= \frac{1}{e-1} \int_0^1 x e^x dx$$

$u = x \quad dv = e^x dx$
 $du = dx \quad v = e^x$

$$= \frac{1}{e-1} [x e^x - \int e^x dx]_0^1$$

$$= \frac{1}{e-1} [x e^x - e^x]_0^1 = \frac{1}{e-1} [(e - e) - (0 - 1)] = \boxed{\frac{1}{e-1}}$$

$$\bar{y} = \frac{1}{2A} \int_0^1 [(e^x)^2 - 0^2] dx$$

$$= \frac{1}{2(e-1)} \left[\frac{e^{2x}}{2} \right]_0^1 = \frac{1}{4(e-1)} [e^2 - 1] = \frac{(e-1)(e+1)}{4(e-1)}$$

$$\Rightarrow P(\bar{x}, \bar{y}) = P\left(\frac{1}{e-1}, \frac{e+1}{4}\right)$$

$$y = \frac{1}{x^2 - 2x + 5} \quad , \quad y = 0, x = -1 \text{ and } x = 3$$

$$\begin{aligned}\bar{x} &= \frac{1}{A} \int_{-1}^3 x \frac{1}{x^2 - 2x + 5} dx \\&= \frac{1}{\frac{7}{2}} \int_{-1}^3 \frac{x}{x^2 - 2x + 5} dx \\&= \frac{2}{7} \int_{-1}^3 \frac{\frac{1}{2}(2x-2) + 1}{x^2 - 2x + 5} dx \\&= \frac{2}{7} \int_{-1}^3 \left(\frac{\frac{1}{2}(2x-2)}{x^2 - 2x + 5} + \frac{1}{x^2 - 2x + 5} \right) dx \\&= \frac{2}{7} \left[\frac{1}{2} \ln|x^2 - 2x + 5| + \frac{1}{2} \tan^{-1}\left(\frac{x-1}{2}\right) \right]_{-1}^3 \\&= \frac{2}{7} \left[\left(\frac{1}{2} \ln|9 - 6 + 5| + \frac{1}{2} \frac{\pi}{4} \right) - \left(\frac{1}{2} \ln|1 + 2 + 5| + 0 \right) \right]\end{aligned}$$

OR
ملفوف
المثلث
 $\frac{\pi}{4}$

$x^2 - 2x + 5$
 $(x-1)^2 - 1 + 5$
 $(x-1)^2 + 4$

= ①

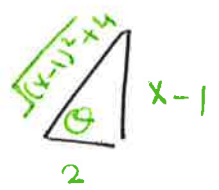
$$\bar{y} = \frac{1}{2A} \int_{-1}^3 \left(\frac{1}{x^2 - 2x + 5} \right)^2 dx$$

$$= \frac{1}{2 \cdot \frac{\pi}{4}} \int_{-1}^3 \frac{1}{(x^2 - 2x + 5)^2} dx \rightarrow \text{التكامل}$$

$$(x-1)^2 - 1 + 5 = (x-1)^2 + 4$$

$$= \frac{1}{\frac{\pi}{2}} \int_{-1}^3 \frac{1}{[(x-1)^2 + 4]^2} dx$$

Let $x-1 = 2\tan\theta \rightarrow \frac{x-1}{2} = \tan\theta$
 $dx = 2\sec^2\theta d\theta$



$$= \frac{\pi/2}{\pi/2} \int_{-1}^3 \frac{1}{[(2\tan\theta)^2 + 4]^2} \cdot 2\sec^2\theta d\theta$$

$$= \frac{\pi/2}{\pi/2} \int_{-1}^3 \frac{1}{[4\tan^2\theta + 4]^2} \cdot 2\sec^2\theta d\theta$$

$$4\tan^2\theta + 4 = 4(\tan^2\theta + 1)$$

$$4\sec^2\theta$$

$$= \frac{\pi/2}{\pi/2} \int_{-1}^3 \frac{1}{16\sec^4\theta} \cdot 2\sec^2\theta d\theta = \frac{2}{\pi} \int_{-1}^3 \frac{1}{8\sec^2\theta} d\theta$$

$$\Rightarrow \frac{1}{\pi} \cdot \frac{1}{4} \int \cos^2\theta d\theta$$

$$= \frac{1}{4\pi} \cdot \frac{1}{2} \int (1 + \cos 2\theta) d\theta$$

$$= \frac{1}{8\pi} \left[\theta + \frac{\sin 2\theta}{2} \right] = \frac{1}{8\pi} \left[\theta + \frac{2\sin\theta \cos\theta}{2} \right]$$

$$= \frac{1}{8\pi} \left[\tan^{-1}\left(\frac{x-1}{2}\right) + \frac{x-1}{\sqrt{(x-1)^2 + 4}} \cdot \frac{2}{\sqrt{(x-1)^2 + 4}} \right]_{-1}^3$$

$$= \dots = \frac{\pi + 2}{16\pi}$$

$$\Rightarrow P(\bar{x}, \bar{y}) = \left(1, \frac{\pi + 2}{16\pi} \right)$$

Find the y-coordinate centroid of the region bounded by the curves

$$y = \frac{1}{x^2 + 1} , \quad y = \frac{1}{x + 1}$$

$$y = y$$

$$\frac{1}{x^2 + 1} = \frac{1}{x + 1}$$

$$x + 1 = x^2 + 1$$

$$x + 1 - x^2 - 1 = 0 \rightarrow x - x^2 = 0 \rightarrow x(1 - x) = 0$$

$$\Rightarrow x = 0 , \quad x = 1$$

at $x = \frac{1}{2}$ $\begin{cases} y_1 = \frac{1}{\frac{1}{4} + 1} = \frac{1}{\frac{5}{4}} = \frac{4}{5} \\ y_2 = \frac{1}{\frac{1}{2} + 1} = \frac{1}{\frac{3}{2}} = \frac{2}{3} \end{cases}$

أو حد المقام $\frac{12}{15} = \frac{4}{5}$
 $\frac{10}{15} = \frac{2}{3}$ $\rightarrow y_1, y_2$

$$\begin{aligned} A &= \int_0^1 \left[\frac{1}{x^2 + 1} - \frac{1}{x + 1} \right] dx \\ &= \left[\tan^{-1} x - \ln|x + 1| \right]_0^1 \\ &= \frac{\pi}{4} - \ln 2 - 0 = \frac{\pi - 4\ln 2}{4} \end{aligned}$$

$$\begin{aligned} \bar{y} &= \frac{1}{2A} \int_0^1 [y_1^2 - y_2^2] dx \\ &= \frac{1}{\frac{\pi - 4\ln 2}{4}} \int_0^1 \left[\frac{1}{(x^2 + 1)^2} - \frac{1}{(x + 1)^2} \right] dx \end{aligned}$$

$$y = \frac{4}{\pi - 4 \ln 2} \int_0^1 \left[\frac{1}{(x^2+1)^2} - (x+1)^{-2} \right] dx$$

$\swarrow$ ↘
 اللاحق $(x+1)^{-1}$
 $\frac{-1}{-1}$

$$\int_0^1 \frac{1}{(x^2+1)^2} dx$$

$$\left[\frac{-1}{x+1} \right]_0^1$$

$$-\frac{1}{2} + 1 = \frac{1}{2}$$

Let $x = \tan \theta$
 $dx = \sec^2 \theta d\theta$



$$\int \frac{1}{(\tan^2 \theta + 1)^2} \cdot \sec^2 \theta d\theta$$

$$\rightarrow (\sec^2 \theta)^2 = \sec^4 \theta$$

$$\int \frac{1}{\sec^4 \theta} \sec^2 \theta d\theta = \int \frac{1}{\sec^2 \theta} d\theta$$

$$\int \cos^2 \theta d\theta = \frac{1}{2} \int [1 + \cos 2\theta] d\theta$$

$$= \frac{1}{2} \left[\theta + \frac{\sin 2\theta}{2} \right] = \frac{1}{2} \left[\theta + \frac{2 \sin \theta \cos \theta}{2} \right]$$

$$= \frac{1}{2} \left[\tan^{-1} x + \frac{x}{\sqrt{x^2+1}} \cdot \frac{1}{\sqrt{x^2+1}} \right]_0^1$$

$$= \frac{1}{2} \left[\frac{\pi}{4} + \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} - 0 \right] = \frac{1}{2} \left[\frac{\pi}{4} + \frac{1}{2} \right]$$

$$= \frac{1}{2} \left[\frac{\pi}{4} + \frac{2}{4} \right] = \frac{1}{2} \frac{\pi+2}{(4)} = \frac{\pi+2}{8}$$

Find the centroid of the region bounded by

$$x = 5 - y^2 \quad \text{and} \quad x = 0$$

$$y^2 = 5 - x$$

$$y_1 = \sqrt{5 - x} \quad y_2 = -\sqrt{5 - x} \quad x = 0$$

$$y = y$$

$$\sqrt{5 - x} = -\sqrt{5 - x}$$

$$\begin{aligned} 2\sqrt{5 - x} &= 0 \\ \div 2 \quad \sqrt{5 - x} &= 0 \longrightarrow 5 - x = 0 \longrightarrow x = 5 \end{aligned}$$

$$A = \int_0^5 [\sqrt{5 - x} - (-\sqrt{5 - x})] dx$$

$$= \int_0^5 2\sqrt{5 - x} dx$$

$$= 2 \left[\frac{(5 - x)^{\frac{3}{2}}}{(-1)\frac{3}{2}} \right]_0^5$$

$$= 2 \left(-\frac{2}{3} \right) [0 - 5^{\frac{3}{2}}] = \frac{4(5^{\frac{3}{2}})}{3}$$

$$\bar{x} = \frac{1}{A} \int_0^5 x (\sqrt{5-x} - (-\sqrt{5-x})) dx$$

$$= \frac{3}{4(5)^{\frac{3}{2}}} \int_0^5 x \cdot 2\sqrt{5-x} dx$$

$$= \frac{3(2)}{4(5)^{\frac{3}{2}}} \int_0^5 x \sqrt{5-x} dx$$

$$y = \sqrt{5-x}$$

$$y^2 = 5-x$$

$$x = 5 - y^2$$

$$dx = -2y dy$$

$$= \frac{3}{2(5)^{\frac{3}{2}}} \int (5-y^2) y \cdot 2y dy$$

$$= \frac{3(2)}{2(5)^{\frac{3}{2}}} \int (5y^2 - y^4) dy$$

$$= \frac{3}{5^{\frac{3}{2}}} \left[5 \frac{y^3}{3} - \frac{y^5}{5} \right] = \frac{3}{5^{\frac{3}{2}}} \left[\frac{5}{3} (\sqrt{5-x})^3 - \frac{(\sqrt{5-x})^5}{5} \right]_0^5$$

$$= \frac{3}{5^{\frac{3}{2}}} \left[0 - \left(\frac{5}{3} (15)^{\frac{3}{2}} - \frac{(15)^{\frac{5}{2}}}{5} \right) \right] = \frac{3}{5^{\frac{3}{2}}} \left(\frac{5(15)^{\frac{3}{2}}}{3} - \frac{(15)^{\frac{5}{2}}}{5} \right)$$

$$C_1 = \frac{1}{2A} \int_0^5 [(\sqrt{5-x})^2 - (-\sqrt{5-x})^2] dx = 4$$

$$= 0$$

Find the centroid of the region bounded by

$$y = x^2 \quad \text{and} \quad x = y^2$$

$$y = x^2$$

$$y = \pm \sqrt{x}$$

$$y = y$$

$$x^2 = \pm \sqrt{x}$$

$$\Rightarrow y_1 = x^2$$

$$y_2 = \sqrt{x}$$

$$y = y$$

$$x^2 = \sqrt{x}$$

$$x^4 = x$$

$$x^4 - x = 0$$

$$\rightarrow x(x^3 - 1) = 0$$

$$x = 0$$

$$x^3 - 1 = 0 \rightarrow x = 1$$

at $x = \frac{1}{4}$ $\rightarrow y_1 = \left(\frac{1}{4}\right)^2 = \frac{1}{16}$
 $y_2 = \sqrt{\frac{1}{4}} = \frac{1}{2} \rightarrow y_2 > y_1$

$$A = \int_0^1 (\sqrt{x} - x^2) dx = \left[\frac{2}{3} x^{\frac{3}{2}} - \frac{x^3}{3} \right]_0^1$$

$$= \frac{2}{3} - \frac{1}{3} = \frac{1}{3}$$

$$\bar{x} = \frac{1}{A} \int_0^1 x (\sqrt{x} - x^2) dx$$

$$= \frac{1}{\frac{1}{3}} \int_0^1 (x^{\frac{3}{2}} - x^3) dx = 3 \left[\frac{2}{5} x^{\frac{5}{2}} - \frac{x^4}{4} \right]_0^1$$

$$= 3 \left[\frac{2}{5} - \frac{1}{4} \right] = 3 \left[\frac{8}{20} - \frac{5}{20} \right] = 3 \cdot \frac{3}{20} = \frac{9}{20}$$

$$\bar{y} = \frac{1}{2A} \int_0^1 ((\sqrt{x})^2 - (x^2)^2) dx$$

$$= \frac{1}{2 \cdot \frac{1}{3}} \int_0^1 (x - x^4) dx$$

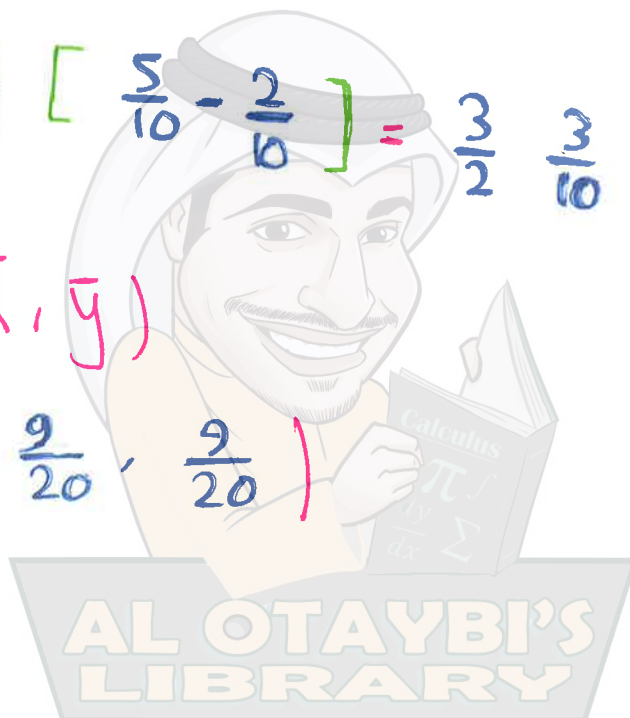
$$= \frac{3}{2} \left[\frac{x^2}{2} - \frac{x^5}{5} \right]_0^1$$

$$= \frac{3}{2} \left[\frac{1}{2} - \frac{1}{5} - 0 \right]$$

$$= \frac{3}{2} \left[\frac{5}{10} - \frac{2}{10} \right] = \frac{3}{2} \cdot \frac{3}{10} = \frac{9}{20}$$

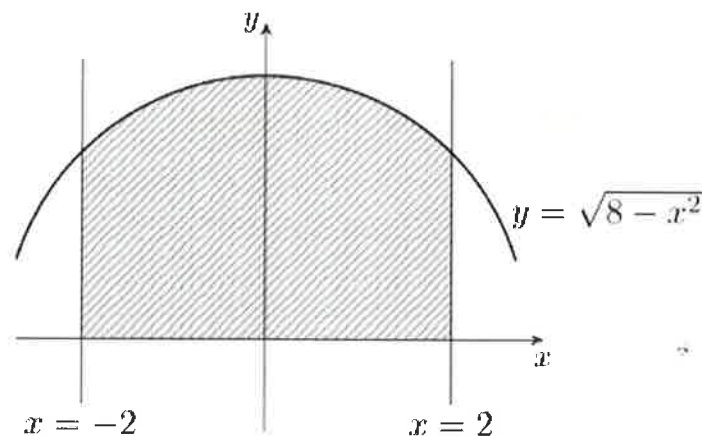
$$\Rightarrow P(\bar{x}, \bar{y})$$

$$= P\left(\frac{9}{20}, \frac{9}{20}\right)$$



Find the centroid of the shaded region.

May 2017

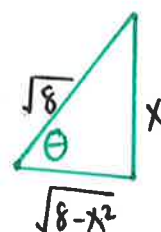


From the region $\rightarrow \bar{x} = 0$

$$A = \int_{-2}^2 [\sqrt{8-x^2} - 0] dx$$

Let $x = \sqrt{8} \sin \theta \rightarrow \sin \theta = \frac{x}{\sqrt{8}}$

$$dx = \sqrt{8} \cos \theta d\theta$$



$$= \int \sqrt{8-8\sin^2\theta} \sqrt{8} \cos \theta d\theta$$

$$\sqrt{8(1-\sin^2\theta)} = \sqrt{8\cos^2\theta} = \sqrt{8} \cos \theta$$

$$= \int \sqrt{8} \cos \theta \sqrt{8} \cos \theta d\theta$$

$$= 8 \int \cos^2 \theta d\theta = 8 \cdot \frac{1}{2} \int (1 + \cos 2\theta) d\theta$$

$$= 4 \left[\theta + \frac{\sin 2\theta}{2} \right] = 4 \left[\theta + \frac{2\sin \theta \cos \theta}{2} \right]$$

$$= 4 \left[\sin^{-1}\left(\frac{x}{\sqrt{8}}\right) + \frac{x}{\sqrt{8}} \cdot \frac{\sqrt{8-x^2}}{\sqrt{8}} \right]_{-2}^2$$

$$\sqrt{8} = \sqrt{4(2)} = 2\sqrt{2}$$

$$= 4 \left[\left(\sin^{-1}\left(\frac{1}{\sqrt{2}}\right) + \frac{1}{\sqrt{2}} \cdot \frac{2}{2\sqrt{2}} \right) - \left(\sin^{-1}\left(\frac{-1}{\sqrt{2}}\right) - \frac{1}{\sqrt{2}} \cdot \frac{2}{2\sqrt{2}} \right) \right]$$

$$= 4 \left[\frac{\pi}{4} + \frac{1}{2} + \frac{\pi}{4} + \frac{1}{2} \right] = 4 \left[\frac{\pi}{2} + 1 \right] = 2\pi + 4$$

$$\bar{y} = \frac{1}{2A} \int_{-2}^2 [(\sqrt{8-x^2})^2 - 0^2] dx$$

$$= \frac{1}{2[2\pi+4]} \int_{-2}^2 [8-x^2] dx$$

$$= \frac{1}{2[2\pi+4]} \left[8x - \frac{x^3}{3} \right]_{-2}^2$$

$$= \frac{1}{2[2\pi+4]} \left[16 - \frac{8}{3} - \left(-16 + \frac{8}{3} \right) \right]$$

$$= \frac{1}{2[2\pi+4]} \left[32 - \frac{16}{3} \right]$$

$$= \frac{1}{2(2)[\pi+2]} \frac{80}{3} = \frac{20}{3\pi+6}$$

$$\Rightarrow \therefore P(\bar{x}, y) = P\left(0, \frac{20}{3\pi+6}\right)$$



Tricky question !!



Find the centroid of the triangle with vertices

$(0, 1)$, $(3, 1)$, $(3, 3)$

The equation of the first line

$$y = 1$$

The equation of the second line

$$x = 3$$

The equation of the third line

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{3 - 1}{3 - 0} = \frac{2}{3}$$

$$y - y_0 = m(x - x_0)$$

$$y - 1 = \frac{2}{3}(x - 0) \rightarrow y - 1 = \frac{2}{3}x \rightarrow y = \frac{2}{3}x + 1$$

Now $y_1 = \frac{2}{3}x + 1$ $y_2 = 1$ $x = 3$

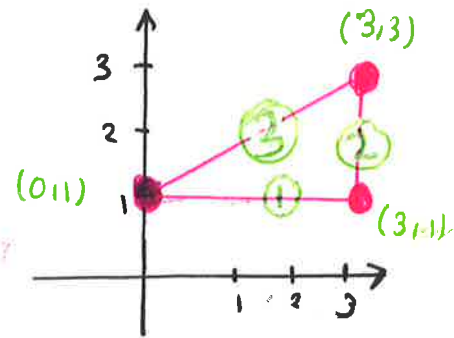
$$y = y$$

$$\frac{2}{3}x + 1 = 1 \rightarrow \frac{2}{3}x = 0 \rightarrow x = 0$$

at $x = 1$ $y_1 = \frac{2}{3} + 1 = \frac{5}{3} \rightarrow y_1, y_2$
 $y_2 = 1$

$$A = \int_0^3 \left(\frac{2}{3}x + 1 - 1 \right) dx = \frac{2}{3} \left(\frac{x^2}{2} \right)_0^3 = \frac{2}{3} \cdot \frac{9}{2} = 3$$

اقترب المثلث من
فساد المثلث

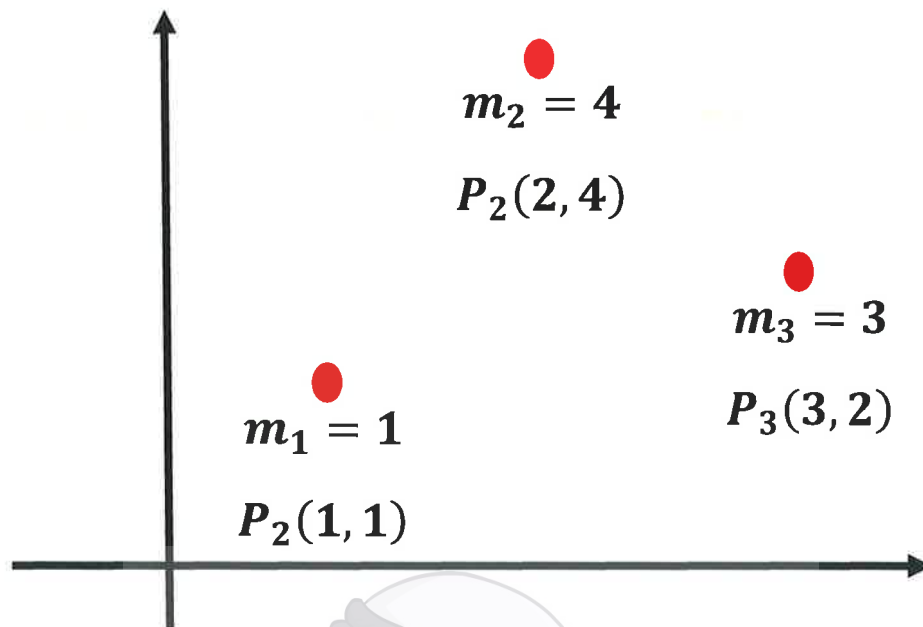


$$\begin{aligned}
 \bar{X} &= \frac{1}{A} \int_0^3 x (y_1 - y_2) dx \\
 &= \frac{1}{3} \int_0^3 x \left(\frac{2}{3}x + 1 - 1 \right) dx \\
 &= \frac{1}{3} \int_0^3 \frac{2}{3} x^2 dx \\
 &= \frac{1}{3} \cdot \frac{2}{3} \left(\frac{x^3}{3} \right)_0^3 = \frac{2}{9} \left(\frac{27}{3} - 0 \right) = 2
 \end{aligned}$$

$$\begin{aligned}
 \bar{y} &= \frac{1}{2A} \int_0^3 [y_1^2 - y_2^2] dx \\
 &= \frac{1}{2(3)} \int_0^3 \left[\left(\frac{2}{3}x + 1 \right)^2 - (1)^2 \right] dx \\
 &= \frac{1}{6} \int_0^3 \left[\frac{4}{9}x^2 + \frac{4}{3}x + 1 - 1 \right] dx \\
 &= \frac{1}{6} \left[\frac{4}{9} \frac{x^3}{3} + \frac{4}{3} \frac{x^2}{2} \right]_0^3 \\
 &= \frac{1}{6} \left[\frac{4}{27} (27) + \frac{2}{3} (9) \right] \\
 &= \frac{1}{6} [4 + 6] = \frac{1}{6} 10 = \frac{5}{3}
 \end{aligned}$$

$$\Rightarrow P(\bar{X}, \bar{y}) = P\left(2, \frac{5}{3}\right)$$

Find the moment and center of mass of the system



$$\begin{aligned} M_x &= \sum m_i y_i \\ &= (1)(1) + 4(4) + 3(2) \\ &= 1 + 16 + 6 = 23 \end{aligned}$$

$$\begin{aligned} M_y &= \sum m_i x_i \\ &= (1)(1) + 4(2) + 3(3) \\ &= 1 + 8 + 9 = 18 \end{aligned}$$

$$\bar{x} = \frac{M_y}{\sum m_i} = \frac{18}{1+4+3} = \frac{18}{8} = \frac{9}{4}$$

$$\bar{y} = \frac{M_x}{\sum m_i} = \frac{23}{1+4+3} = \frac{23}{8}$$

$$\Rightarrow \therefore P(\bar{x}, \bar{y}) = \left(\frac{9}{4}, \frac{23}{8} \right)$$

The masses m_i are located at the point P_i . Find the moments M_x, M_y about the origin, then find the center of mass.

$m_1 = 6$	$m_2 = 5$	$m_3 = 1$	$m_4 = 4$
$P_1(1, -2)$	$P_2(3, 4)$	$P_3(-3, -7)$	$P_4(6, -1)$

$$M_x = \sum m_i y_i$$

$$= 6(-2) + 5(4) + 1(-7) + 4(-1)$$

$$= -12 + 20 - 7 - 4 = -3$$

$$M_y = \sum m_i x_i$$

$$= 6(1) + 5(3) + 1(-3) + 4(6)$$

$$= 6 + 15 - 3 + 24 = 42$$

$$\bar{x} = \frac{M_y}{\sum m_i} = \frac{42}{6+5+1+4} = \frac{42}{16} = \frac{21}{8}$$

$$\bar{y} = \frac{M_x}{\sum m_i} = \frac{-3}{6+5+1+4} = \frac{-3}{16}$$

$$\Rightarrow \therefore P(\bar{x}, \bar{y}) = \left(\frac{21}{8}, \frac{-3}{16} \right)$$

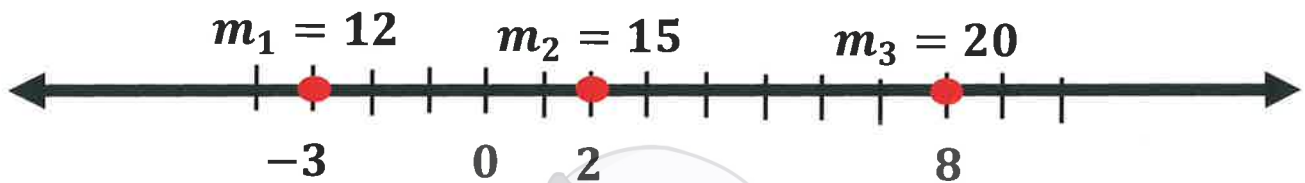


Text book question



Point masses m_i are located on the x - axis as shown.

Find the moment M of the system about the origin ,
then find the center of mass.



$$\begin{aligned} M_x &= \sum m_i y_i \\ &= 12(0) + 15(0) + 20(0) = 0 \end{aligned}$$

لا يوجد بعد رأسي

$$\begin{aligned} M_y &= \sum m_i x_i \\ &= 12(-3) + 15(2) + 20(8) \\ &= -36 + 30 + 160 = 154 \end{aligned}$$

$$\bar{x} = \frac{M_y}{\sum m_i} = \frac{154}{12+15+20} = \frac{154}{47}$$

$$\bar{y} = \frac{M_x}{\sum m_i} = \frac{0}{12+15+20} = 0$$

$$\Rightarrow \therefore P(\bar{x}, \bar{y}) = \left(\frac{154}{47}, 0 \right)$$